RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE NAME										
CLASS	6									
CENTRE NUMBER	S					INDEX NUMBER				

H1 CHEMISTRY 8872/01

Paper 1 Multiple Choice

22 September 2011

50 minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, class, Centre number and index number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **11** printed pages.

Section A

For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider to be correct.

1. Complete combustion of a sample of a hydrocarbon, **Y**, gave 0.66 g of carbon dioxide and 0.36 g of water. The empirical formula of **Y** is

A C_3H_4 **B** C_3H_5 **C** C_3H_8 **D** C_6H_8

2. Which of the following has the greatest mass?

A 6×10^{23} molecules of hydrogen gas

B 3 moles of nitrogen molecules

C 1.2×10^{24} atoms of iron

D 3×10^{23} ions of copper(II)

3. 10 cm^3 of 0.2 mol dm^{-3} K_2**X**O_4 will react with 40 cm^3 of 0.1 mol dm^{-3} iron(II) sulfate solution. If Fe^{2+} is oxidised to Fe^{3+} by K_2**X**O_4, what is the final oxidation state of **X**?

A +2 **B** +3 **C** +4 **D** +5

4. Which one of the following ions has the most number of unpaired electrons?

A Cr^{3+} **B** Ni^{2+} **C** Ca^{2+} **D** Co^{3+}

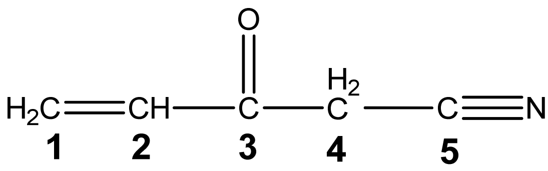
5. Which of the following species contains the smallest bond angle?

A N_2H_4 **B** BrF_5 **C** XeF_2 **D** SO_2

6. Which gas can be most easily liquefied by cooling and applying pressure?

A HF **B** Ne **C** H_2 **D** CH_4

7. Consider the molecule:



What is the hybridisation of each of the carbon atoms, **C1** to **C5**?

	C1	**C2**	**C3**	**C4**	**C5**
A	sp^2	sp^2	sp^2	sp^3	sp
B	sp^2	sp^2	sp^2	sp^3	sp^3
C	sp^2	sp^2	sp^3	sp^3	sp
D	sp^2	sp^2	sp^3	sp^3	sp^3

8. Given the information:

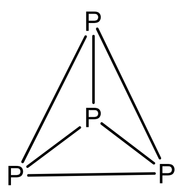
$\Delta H_c(C) = -394$ kJ mol^{-1}
$\Delta H_f(H_2O) = -286$ kJ mol^{-1}
$\Delta H_f(CH_3OH) = -239$ kJ mol^{-1}

Which one of the following is the correct enthalpy change of combustion of methanol, CH_3OH, in kJ mol^{-1}?

A –441 **B** –727 **C** –919 **D** –1205

9. Use of the *Data Booklet* is relevant to this question.

Phosphorus, P_4, has the following molecular structure:



Imagine that nitrogen were to form a similar molecule N_4:

$$2N_2(g) \rightarrow N_4(g)$$

By considering the bonds broken and the bonds formed, what would be the value of ΔH (in kJ mol^{-1}) for the above reaction?

A 1028 **B** 1328 **C** 1954 **D** 2628

10. For which equilibrium does K_c have no units?

A $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$

B $CH_3OH(l) + CH_3CO_2H(l) \rightleftharpoons CH_3CO_2CH_3(l) + H_2O(l)$

C $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

D $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

11. What is the final pH of a solution formed by mixing equal volumes of two separate solutions of pH 2.0 and pH 4.0?

A 2.0 **B** 2.3 **C** 3.0 **D** 3.3

12. Which one of these statements can be deduced from the following information?

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = -98 \text{ kJ mol}^{-1}$$

A Increasing the pressure increases the value of the equilibrium constant.

B Increasing the pressure decreases the proportion of SO_2 and O_2 at equilibrium.

C Increasing the temperature decreases the rate of the forward reaction.

D Adding a catalyst increases the yield and rate of production of SO_3.

13. A substance **P** changes by a first order reaction to form the product **Q**, according to the equation:

$$P \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} Q$$

k_1 = rate constant of the forward reaction.

k_{-1} = rate constant of the backward reaction.

Which of the following expression represents the equilibrium constant, K_c?

A k_1 **B** $k_1[P]$

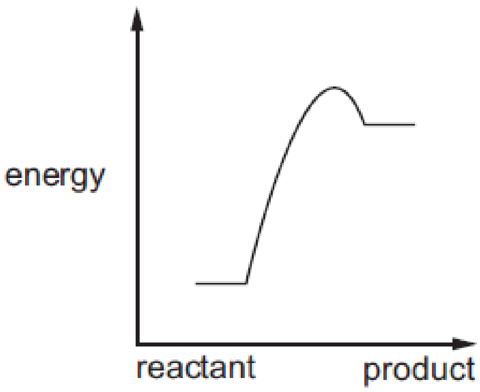
C $\dfrac{k_1}{k_{-1}}$ **D** $\dfrac{k_1[P]}{k_{-1}[Q]}$

14. Four reactions of the type shown are studied at the same temperature.

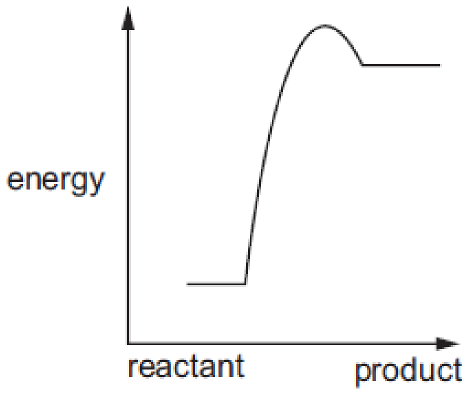
$X(g) + Y(g) \rightleftharpoons Z(g)$

Which is the correct reaction pathway diagram for the reaction that would proceed most rapidly and with good yield?

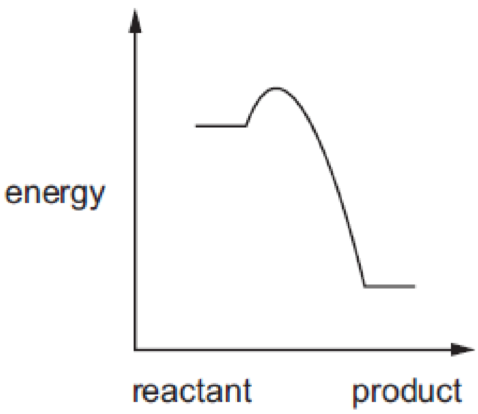
A



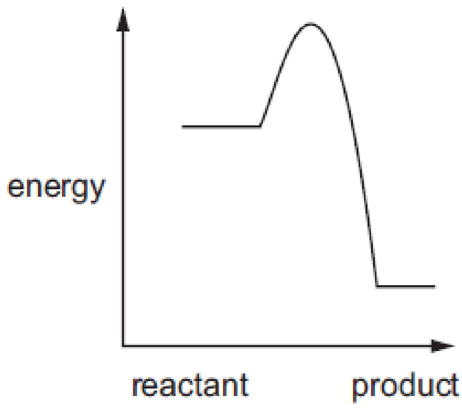
B



C



D



15. Which of the following elements is expected to show the greatest tendency to form covalent compounds?

A Barium

B Calcium

C Magnesium

D Potassium

16. An element **Y** has a low proton number. It forms an acidic oxide which is insoluble in water and a chloride which is readily hydrolysed by water.

Which Group in the Periodic Table might contain **Y**?

A II **B** III **C** IV **D** V

17. Which of the following is the least likely product formed in the termination step in the reaction of CH_3CH_3 with Cl_2 in the presence of UV light?

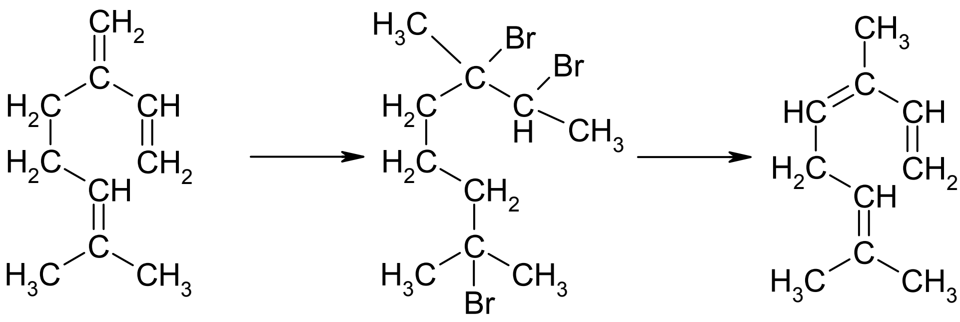
A HC*l*

B CH_3CH_2Cl

C CH_3CHCl_2

D C_2H_6

18. *Myrcene* is a chemical produced by pine trees in North America, which attracts females of the beetle *Dendroctonus brevicomins*. This will indirectly cause the death of the tree. To prevent this, *Myrcene* can be converted to its isomer, **Y** via intermediate **X**.



Myrcene **X** **Y**

Which of the following statements is true?

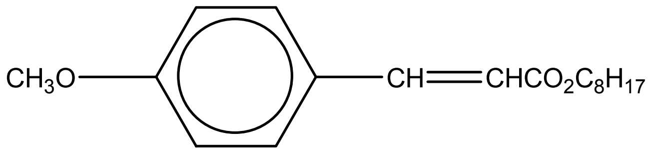
A *Myrcene* reacts with 1.5 moles of bromine to form intermediate **X**.

B *Myrcene* can be distinguished from its isomer **Y** by reaction with hot, acidified $KMnO_4$.

C Intermediate **X** reacts with ethanolic sodium hydroxide to form isomer **Y**.

D Intermediate **X** is formed from *Myrcene* via hydrogenation.

19. The compound 2-ethylhexyl-p-methoxycinnamate (MOC) is used as a sunscreen.

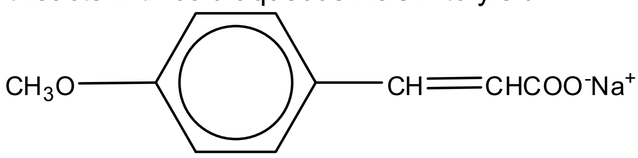


Which of the following statement regarding this compound is correct?

A It is slightly soluble in water.

B A brown precipitate is formed with cold alkaline $KMnO_4$.

C It reacts with cold aqueous NaOH to yield



D A racemic mixture is produced when it is boiled with aqueous hydrochloric acid.

20. Geraniol is found in rose oil and is used for the preparation of artificial scents. Upon controlled oxidation, geraniol yields CH_3COCH_3, $CH_3COCH_2CH_2COOH$ and $H_2C_2O_4$. $H_2C_2O_4$ neutralises completely 2 moles equivalent of NaOH in an acid-base reaction, and can also be oxidised under vigorous conditions to CO_2 and H_2O. A possible structure of geraniol is:

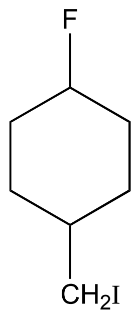
A $CH_3CH{=}C(CH_3)CH_2CH_2C(CH_3){=}CHCH_2OH$

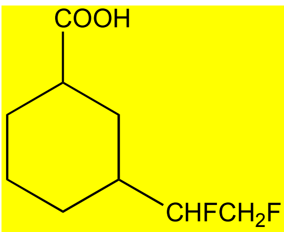
B $(CH_3)_2C{=}CHCH_2CH_2C(CH_3){=}CHCH_2OH$

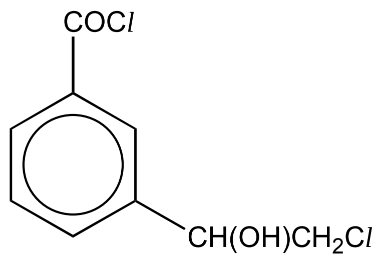
C $(CH_3)_2C{=}C(CH_3)CH_2C(CH_3){=}CHCH_2OH$

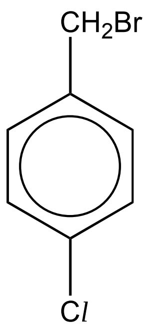
D $(CH_3)_2C{=}CHCH_2CH_2C(CH_2OH){=}CHCH_3$

21. Which of the following compounds will **not** give a precipitate when boiled with aqueous silver nitrate?

A 

B 

C 

D 

22. Concentrated ammonia was heated in a sealed tube with excess bromoethane (C_2H_5Br). Which of the following products will **not** form?

A $C_4H_{11}N$

B $C_5H_{13}N$

C $C_6H_{15}N$

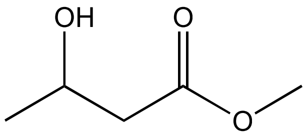
D $C_8H_{20}NBr$

23. Organic compound **P** has the following properties:

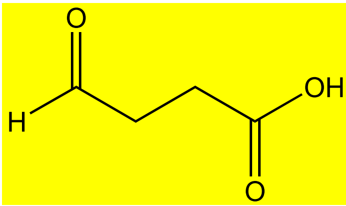
- Produces white fumes with PCl_5
- Forms a brick-red precipitate with Fehling's reagent

Which of the following compounds could be **P**?

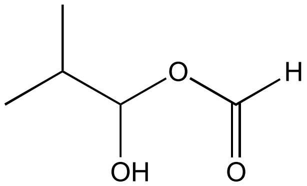
A



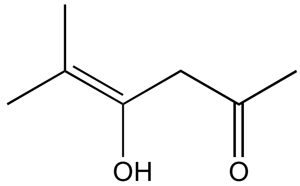
B



C



D



24. Which sequence shows the correct order of increasing ease of hydrolysis?

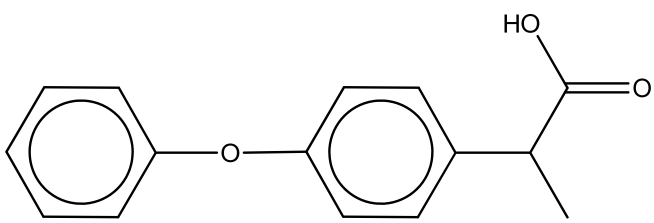
A CH_3CH_2I < C_6H_5Br < $C_6H_5CH_2Cl$

B CH_3CH_2I < $C_6H_5CH_2Br$ < C_6H_5Cl

C $C_6H_5CH_2Br$ < C_6H_5Cl < CH_3CH_2I

D C_6H_5Br < CH_3CH_2Cl < $C_6H_5CH_2I$

25. The anti-arthritic agent *Fenoprofen* has the following structure:



Which of the following reagents show an observable change with *Fenoprofen*?

A 2,4-dinitrophenylhydrazine

B Fehling's reagent

C hot acidified potassium dichromate(VI)

D hot acidified potassium manganate(VII)

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26. Which of the molecules are polar?

1 $C_6H_5CH_3$ **2** CH_3OCH_3 **3** SF_4

27. In which pair of molecules is the boiling point of molecule **II** greater than that of molecule **I**?

	Molecule **I**	Molecule **II**
1	PH_3	NH_3
2	SO_3	SO_2
3	$CH_3CH(CH_3)CH_3$	$CH_3CH_2CH_2CH_3$

28. Which statements about the properties of a catalyst are correct?

1 A catalyst increases the average kinetic energy of the reacting particles.

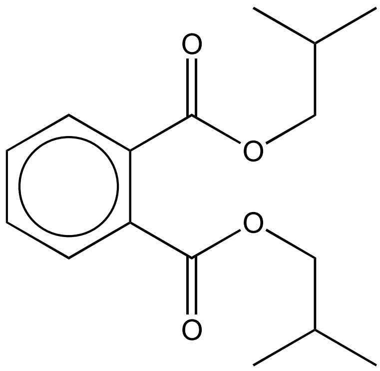
2 A catalyst increases the rate of the reverse reaction.

3 A catalyst has no effect on the enthalpy change ΔH.

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

29. In June 2011, a variety of Taiwanese food products were found to contain diisobutyl phthalate (DIBP), a plasticiser.



Diisobutyl phthalate (DIBP)

Which of the following statements about DIBP are **incorrect**?

1 DIBP produces a pale yellow precipitate when heated with alkaline aqueous iodine.

2 When DIBP is heated with acidified potassium manganate(VII), the purple solution decolourises.

3 When DIBP is heated with dilute sulfuric acid, one of the products requires two molar equivalents of aqueous sodium hydroxide for complete neutralisation.

30. Deuterium, D, is the ${}^{2}_{1}H$ isotope of hydrogen.

Which of the following reactions yield a carbon compound containing deuterium?

1 $CH_2{=}CH_2 \xrightarrow{DCl}$

2 $CH_3CN \xrightarrow[D_2O]{D_2SO_4,\ heat}$

3 $CH_3CH_2OH \xrightarrow{I_2,\ NaOD,\ heat}$